VEDANT LANJEKAR

AI Software Engineer | Applied AI | Generative AI | Full-Stack Development

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PROFESSIONAL SUMMARY

- B.E. student in Artificial Intelligence and Data Science (2023–2027) with SGPA between 9.14 and 9.86 in all six completed semesters; AI Engineer Intern at Valmet Technologies since June 2026.
- Builds AI-powered, full-stack applications end to end, with hands-on work in LLMs and Retrieval-Augmented Generation (RAG), YOLOv8 computer vision, FastAPI backends, local LLM execution and Edge AI.
- Conference research on LLM hallucination detection; winner of national and international hackathons, project exhibitions and research presentations.

CORE TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C, C++, SQL
- **AI / Machine Learning:** TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv8, Computer Vision, Deep Learning, Natural Language Processing (NLP)
- **Generative AI:** Large Language Models (LLMs), RAG, Vector Databases, Transformer Embeddings, LangChain, LangGraph, AI Agents, Ollama, OpenAI API, Gemini API
- **Backend / APIs:** FastAPI, Node.js, Express.js, Flask, Django, REST APIs, WebSockets
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Supabase, Firebase
- **Cloud / DevOps:** Docker, Git, GitHub, GitHub Actions, CI/CD, Vercel, Linux
- **Embedded / IoT:** Embedded C, Arduino, ESP32, Raspberry Pi, MQTT, LoRa
- **Engineering Practices:** Data Structures and Algorithms, Object-Oriented Programming, System Design, API Development, Database Design

PROFESSIONAL EXPERIENCE

AI Engineer Intern — **June 2026 – Present**

Valmet Technologies Private Limited | Baner, Pune, India | On-site

- Developed **eight internal engineering applications from scratch** for an industrial automation business line, from requirements analysis to architecture, implementation and delivery.
- Designed application architectures and technical workflows; implemented engineering-specific business logic to reduce manual effort.
- Built frontend, backend, data-processing, integration and AI-related components across applications for control-system migration, project control, material synchronisation, hardware tracking and workload balancing.

Web Developer Intern — **January 2026 – March 2026**

Bharat Electrical & Engineering | Panvel, Navi Mumbai, India | On-site

- Developed the corporate website, structuring the electrical, HVAC, fire-protection and solar service portfolio for online presentation.
- Implemented responsive layouts and frontend functionality to business requirements; supported maintenance and optimisation of performance, security and scalability.

AI Engineer Intern (training and internship programme) — **February 2024 – April 2024**

Acmegrade x Mood Indigo, IIT Bombay | Online

- Completed structured AI training with applied exercises in AI concepts, workflows and implementation.

Social Media Manager, Photographer and Cinematographer — **January 2024 – June 2026**

Dr. D. Y. Patil College of Engineering & Innovation, Institute of Management and Entrepreneurship, Technical Campus | Pune

- Managed YouTube, Instagram and LinkedIn channels and produced event photography and edited video across three institutions.

EDUCATION

Bachelor of Engineering, Artificial Intelligence and Data Science — **2023 – 2027 (Currently pursuing)**

Dr. D. Y. Patil College of Engineering & Innovation | Pune, India

- **SGPA:** Semester I 9.59 | II 9.27 | III 9.86 | IV 9.32 | V 9.14 | VI 9.73
- **HSC** (2022–2023), CBSE, 73.92% | **SSC** (2021–2022), CBSE, 87.92%, New Horizon Public School, New Panvel
- **Languages:** English (Professional Working), Hindi and Marathi (Native / Full Professional)

SELECTED TECHNICAL PROJECTS

AIVORA Labs — Integrated AI Engineering Ecosystem **November 2025 – Present**

Founder, Lead System Architect and Full-Stack AI Developer | Ideation-stage venture

- Architected a seven-application ecosystem: component knowledge hub (Atlas), IoT simulator (DigiLab), AI solution builder (Forge), AI PCB builder (Flux), engineering community (Nexus), AI CV and portfolio builder (Signature) and assessment platform (Horizon).
- Stack: Python, Node.js, React.js, Vercel. Platform | GitHub

ACE EYE — Real-Time Playing-Card Detection **January 2025 – May 2025**

- *AI Model Developer and Backend Engineer.* Built a YOLOv8 detection and classification model with a FastAPI and WebSockets backend for real-time, GPU-oriented inference. GitHub

VISION BENCH AI — Object-Detection Benchmarking **January 2025 – May 2025**

- *AI Systems Architect and Computer Vision Engineer.* Designed a unified workflow to compare YOLO, SSD and Faster R-CNN on FPS, latency, precision, recall and efficiency (Python, OpenCV). GitHub

Insight Sphere AI — Local Conversational AI **January 2025**

- *AI Backend and Full-Stack Developer.* Built a privacy-oriented chat system running LLMs locally through Ollama, with a modular FastAPI backend and streaming interface. GitHub

ARC-AI — Autonomous Regional Cognitive AI (concept) **January 2025 – May 2025**

- *Embedded AI Systems Developer.* Designed a decentralised offline AI mesh with Main Hub and Mini Hub architecture, local inference, LoRa links and cryptographically signed Knowledge Capsules. GitHub
- **Also built:** VIRO-AI (research concept), VisionDeck Pro, SENTIVIEW, MEDLOCUS, Skillectra, SRANS, BRAILLEAR, Gesture Drive XT, DriveU, SECUREWEB LAB, PROLINK HUB, INSIGHTX, CLIMETRYX.

RESEARCH AND PUBLICATIONS

- **A Hybrid Framework for Hallucination Detection and Mitigation in Large Language Models** (International conference paper; ISSN 2348-6953, Vol. 13, Issue 2s, 2026, pp. 396–408): combined RAG, vector databases, transformer embeddings, semantic similarity and confidence scoring to flag unsupported LLM outputs; methods include TruthfulQA and FEVER.
- **Aerial Pharmacobotany — Drone-Integrated Pharmacobotanical Intelligence Platform** (International conference paper; ISSN 2319-2526, Vol. 14, Issue 01, 2025): UAV imaging, OpenCV, YOLO deep learning and GIS mapping for medicinal plant classification.

CERTIFICATIONS AND TRAINING

- **Certifications:** NPTEL Python for Data Science (IIT Madras); Microsoft Data Analyst Career Path; Cisco Networking Academy: Introduction to Modern AI, Introduction to Data Science; Infosys Springboard: Drone Technology, Traffic Light System (Centrado Kit).
- **Workshops and training (participation):** National-Level DevOps Workshop; One-Week FDP on Cyber-Physical Systems and IoT Application Development; Industrial Training in IoT Do-It-Yourself IDE; VISHWAPRENEUR'25 National-Level Entrepreneurship Convention.

ACHIEVEMENTS AND RECOGNITION

- **International level:** Winner, BV Tech Fusion 2K26 (AIVORA Labs); Research Presentation Winner, International Conference 2K26 (LLM hallucination research); Winner, International Conference 2K25 (Aerial Pharmacobotany); Top 20 Finalist, Fusion 2K26.
- **National winner:** Pragyantra AI Hackathon 2K26 (AIVORA Labs); Sci Tech Villagethon; KIT Ideathon 2K25.
- **National runner-up and podium:** Byte Battle Hackathon 2K26 (Runner-Up); M-Pulse Ignite Design X 2K26 (Runner-Up); VishwaNova Project Exhibition 2K26 (3rd Runner-Up).
- **National finalist:** Orchestron Agentic AI Hackathon 2K26 (Top 5); Ideathon 2K26 and Techno Sci 2K25 (Top 10); Innovators Hackathon 2K26, Fusion Hackathon 2K25 and SciTech Innovation Hackathon 2K25 (Top 20).
- **Institutional and other:** Winner, Pixel War 2K26 UI/UX Design Battle, Idea Pitch Competition (Startup), Internal Hackathon for Smart India Hackathon 2024 and Quick Heal Cybersecurity Poster Competition; Runner-Up, Smart India Hackathon Round 1 (institutional round).

LEADERSHIP AND ENTREPRENEURSHIP

- **President,** Society of Data Science (S4DS) Student Chapter (2026–2027); **Assistant Documentation Secretary,** S4DS (2023–2025).
- **Media / Marketing / Magazine Secretary,** Student Council, DYPCOEI (2023–2025); **Media Head,** Young Inspirators Network Club and Social Media Club (2024–2025).
- **Founder,** AIVORA Labs (ideation stage); team member in ideation-stage initiatives BuildWyse and Stack Notes.